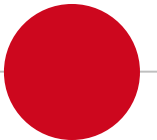


What do we know about *Aedes koreicus* vector competence ?

Federica Gobbo, DVM

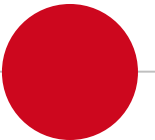
Istituto Zooprofilattico Sperimentale delle Venezie

Laboratorio di entomologia sanitaria e patogeni trasmessi da vettori



What is vector competence vs vector capacity

Why is it important to study vector competence in *Aedes koreicus*



What is Vector Competence?

Vector competence is the **intrinsic** and biological **ability** of an arthropod **vector** (mosquito, tick sandfly) to acquire, maintain, and **transmit** a **pathogen** to a new **host**



OPEN ACCESS

EDITED BY
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Johns Hopkins University, United States

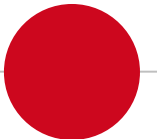
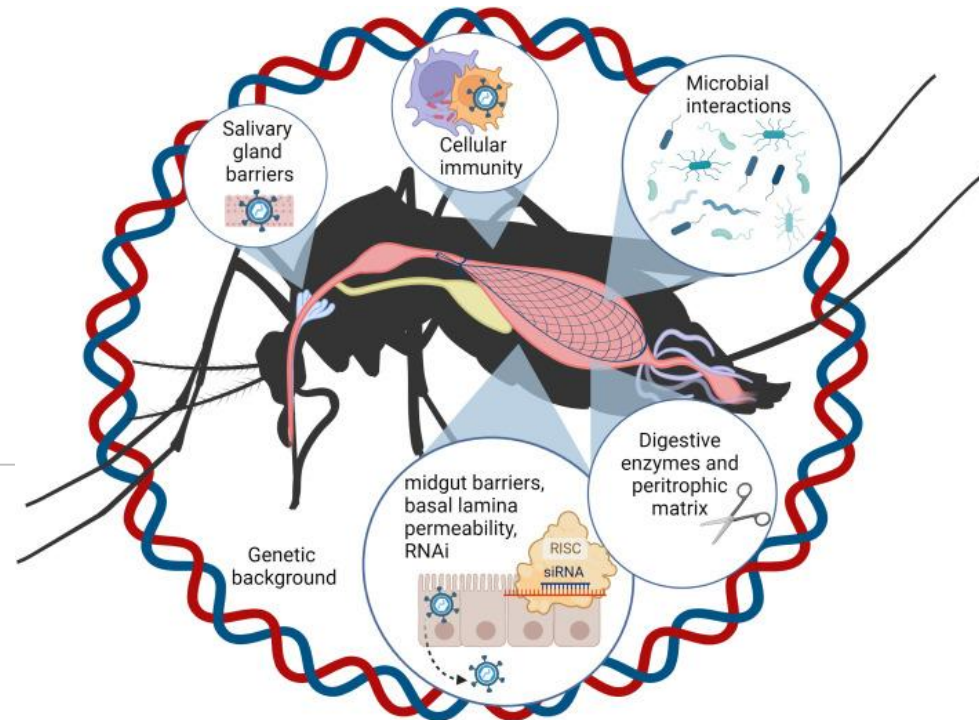
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Intrinsic factors driving mosquito vector competence and viral evolution: a review

Juliette Lewis^{1†}, Emily N. Gallichotte^{1†}, Jenna Randall¹,
Arielle Glass², Brian D. Foy¹, Gregory D. Ebel¹
and Rebekah C. Kading^{1*}

¹Center for Vector-borne Infectious Diseases, Department of Microbiology, Immunology, and Pathology, Colorado State University, Fort Collins, CO, United States, ²Department of Cellular and Molecular Biology, Colorado State University, Fort Collins, CO, United States

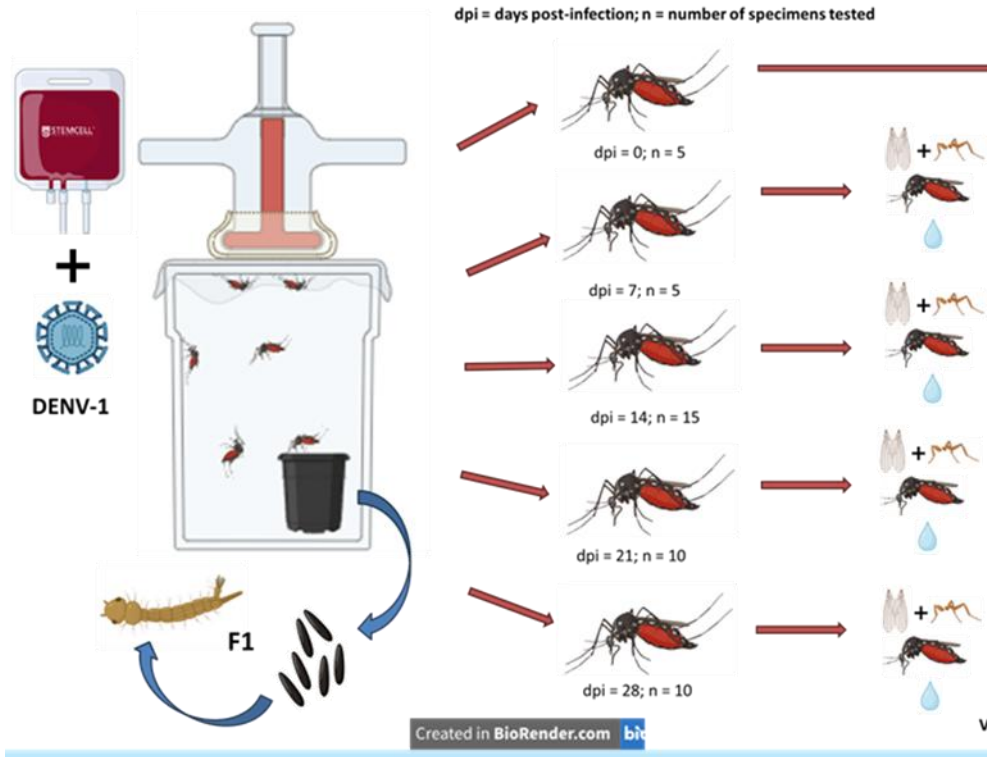


How to perform study of Vector Competence?

- ❑ The calculation of Vector Competence is assed under experimental conditions and in specific Lab. facilities (BSL3)
- ❑ Vectors such as mosquitoes can be infected *under* different routes of administrations, different climatic conditions (T°C, Humidity (%), photoperiods L:D) and different viral load of pathogen



How calculate Vector Competence?



❑ **Infection Rate (IR):** The proportion of mosquitoes with a pathogen positive **midgut (or body)** out of the total number that consumed an infectious blood meal

❑ **Dissemination Rate (DR):** The proportion of infected vectors where the pathogen has spread beyond the midgut to secondary tissues like **legs, wings, or head**

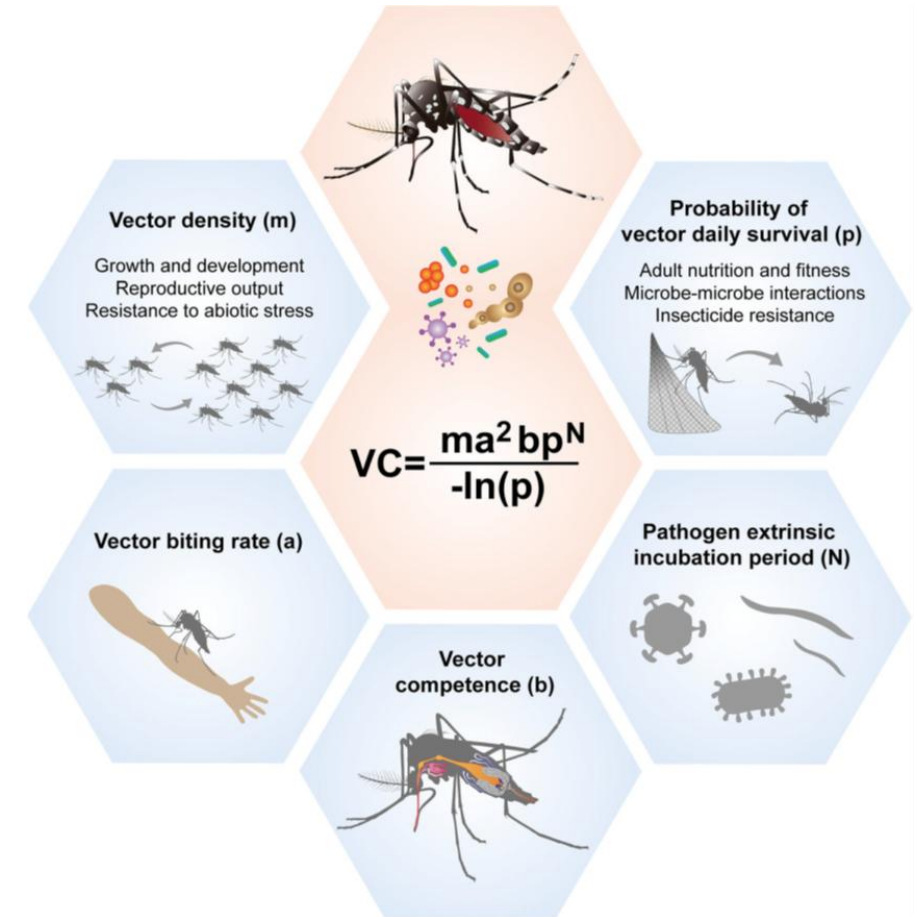
❑ **Transmission Rate (TR):** The proportion of vectors with disseminated infections that have the pathogen present in their **saliva**

❑ **Transmission Efficiency (TE):** "Population Transmission Rate," this measures the total percentage of the initial group capable of transmitting the pathogen

What is Vectorial Capacity (VC)?

Vectorial Capacity (VC) is an equation to model the spread of a pathogen in the field considering external factors like:

- ❑ **(m) Mosquito density:** The number of female mosquitoes per host
- ❑ **(a) Biting rates:** The average number of bites on a host per mosquito per day
- ❑ **(b) Vector Competence:** The probability that a vector becomes infected and successfully transmits the pathogen
- ❑ **(p) The biological efficiency Daily Survival Probability:** The probability that a mosquito will survive to the next day
- ❑ **(n) The Extrinsic Incubation Period – EIP:** The time (in days) required for the pathogen to replicate and reach the salivary gland
- ❑ **(-ln (p)) The daily mortality rate:** The natural logarithm of the survival probability



Zika and chikungunya: mosquito-borne viruses in a changing world

Talya Shragai,¹ Blanka Tesla,² Courtney Murdock,^{2,a} and Laura C. Harrington^{1,a}¹Department of Entomology, Cornell University, Ithaca, New York. ²Department of Infectious Diseases and Odum School of Ecology, University of Georgia, Athens, Georgia

Address for correspondence: Laura C. Harrington, Department of Entomology, Cornell University, 3138 Comstock Hall, Ithaca, NY 14850. lch27@cornell.edu

Shragai *et al.*

Global change and vectors of chikungunya and Zika

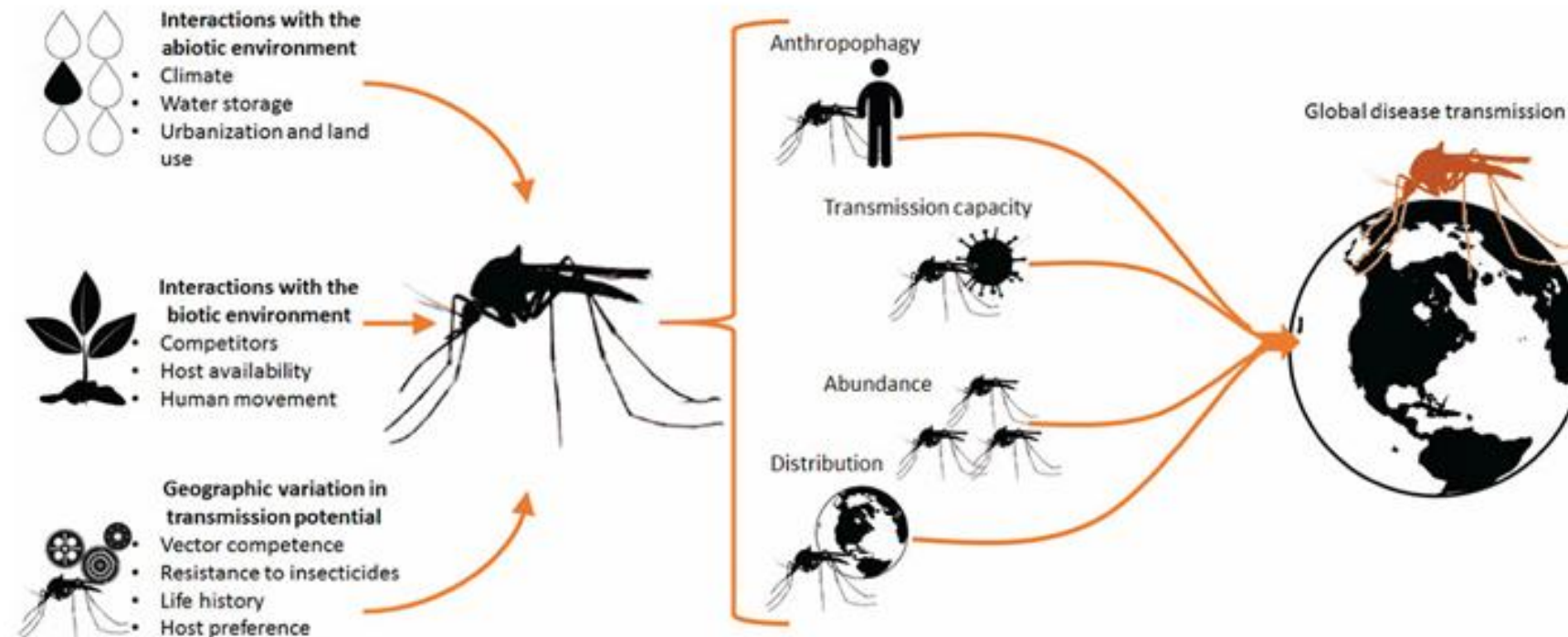
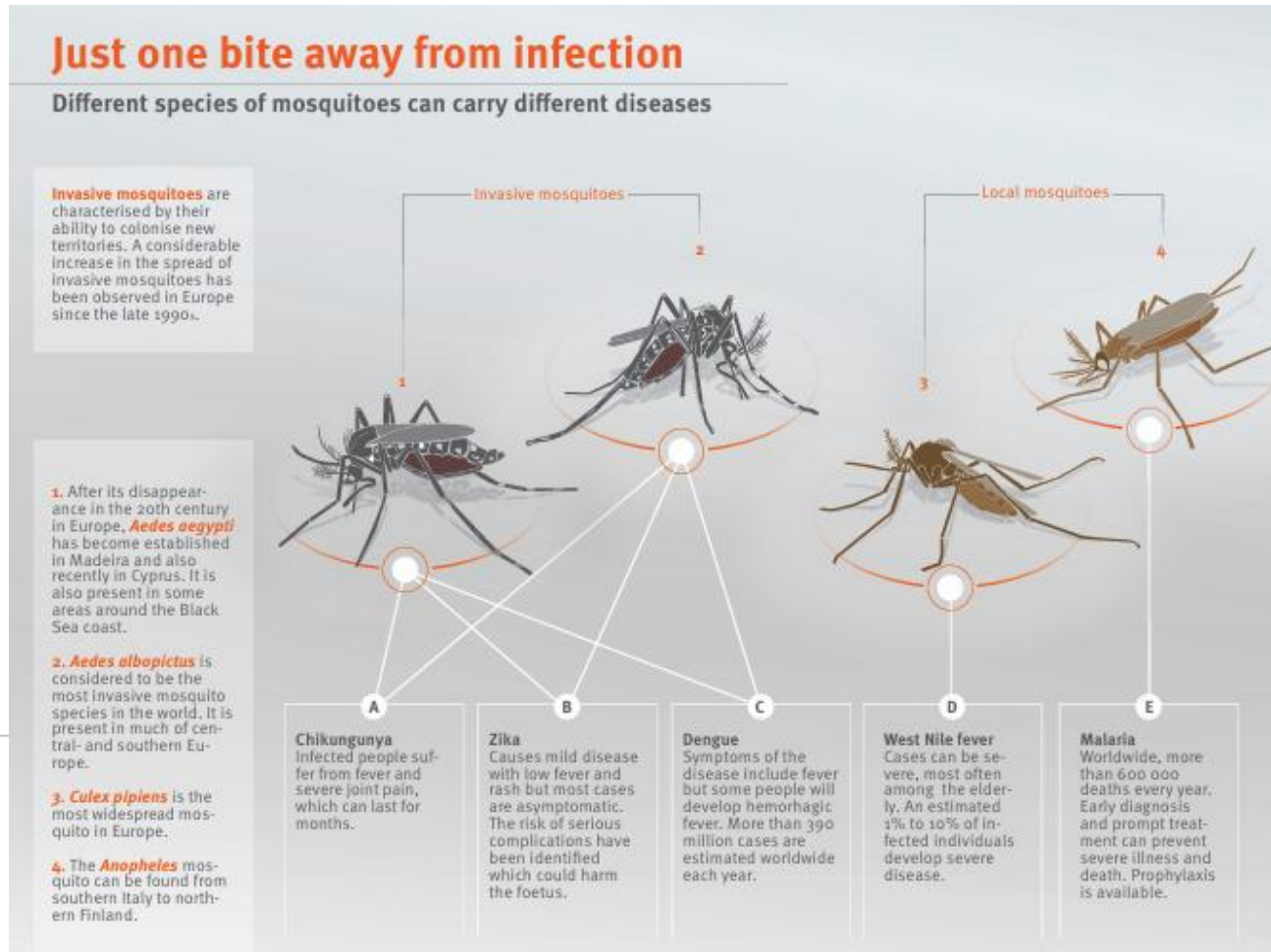
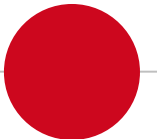


Figure 1. The complexity of interactions among mosquito vectors, arboviral pathogens, and transmission across environmental drivers.

Why is it important to study vector competence in *Aedes koreicus*?

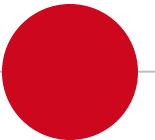


- ☐ Genus *Aedes* spp.
- ☐ Established in Europe and in continuous expansion
- ☐ Urban and anthropophilic habits



Invasive **Aedes** Mosquitoes

<https://www.ecdc.europa.eu/en/publications-data/invasive-mosquitoes-colonising-europe-what-can-we-do>



What we know from literature about vector competence in *Aedes koreicus*?

Bull. Org. mond. Santé } 1964, 30, 197-210
Bull. Wld Hlth Org.

Some Ecological Aspects of the Problem of Arthropod-borne Animal Viruses in the Western Pacific and South-East Asia Regions*

J. A. R. MILES¹

TABLE 2
MOSQUITOS FOUND CARRYING JAPANESE OR MURRAY VALLEY ENCEPHALITIS VIRUSES IN THE FIELD

Japanese encephalitis	Murray Valley encephalitis
<i>Aedes koreicus</i>	
<i>Aedes togoi</i> ^a	<i>Aedes normanensis</i>
(<i>Aedes albopictus</i>) ^b	
<i>Anopheles hyrcanus</i>	
<i>Anopheles sinensis</i>	
<i>Culex gelidus</i> ^a	<i>Culex annulirostris</i> ^a
<i>Culex pipiens</i>	
<i>Culex tritaeniorhynchus</i> ^a	
<i>Culex vishnui</i> complex ^a	

^a Those mosquitoes which appear to be those epidemiologically most important in various areas.

^b One reported isolation from adults reared from larvae collected near the home of a patient.

RESEARCH

Open Access

Development of *Dirofilaria immitis* within the mosquito *Aedes (Finlaya) koreicus*, a new invasive species for Europe

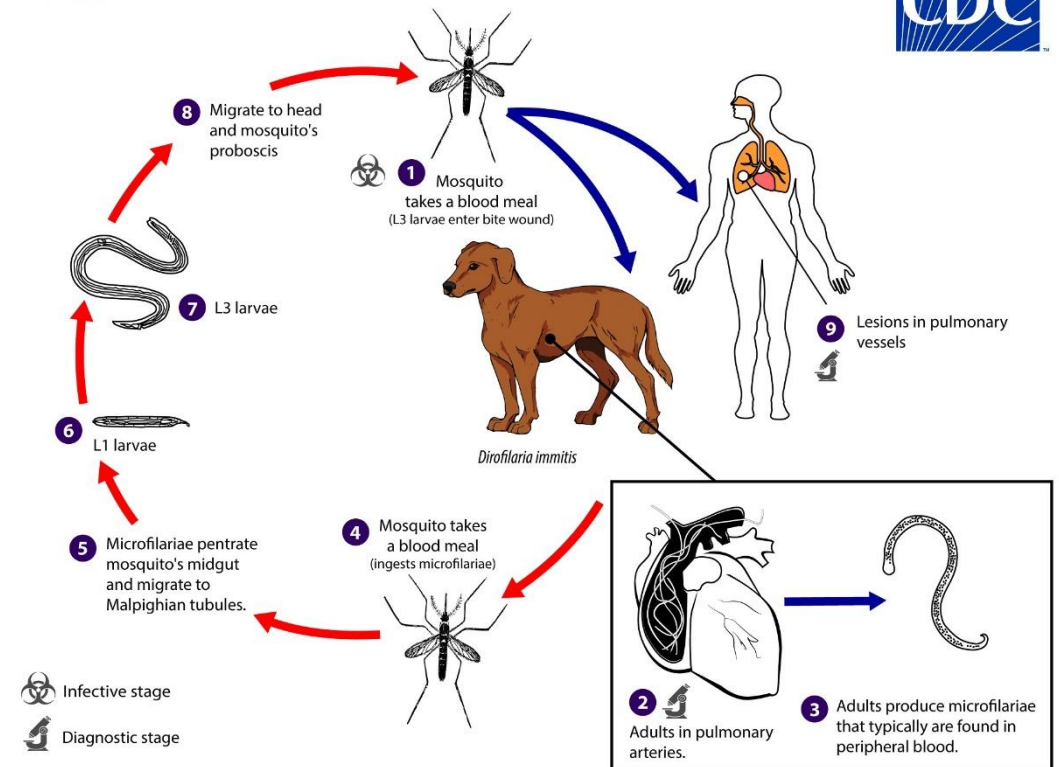
Fabrizio Montarsi¹, Silvia Ciocchetta^{2,3}, Gregor Devine², Silvia Ravagnan¹, Franco Mutinelli¹, Antonio Frangipane di Regalbono⁴, Domenico Otranto⁵ and Gioia Capelli^{1*}



L2 of *Dirofilaria immitis* within the Malpighian tubules of *Aedes koreicus* (5X).

4DPDx

Pulmonary Dirofilariasis



REVIEW



The new European invader *Aedes (Finlaya) koreicus*: a potential vector of chikungunya virus

Silvia Ciocchetta^{a,b} , Natalie A. Prow^a, Jonathan M. Darbro^a, Francesca D. Frentiu^b , Sandro Savino^d, Fabrizio Montarsi^c, Gioia Capelli^c, John G. Aaskov^b and Gregor J. Devine^a

^aQIMR Berghofer Medical Research Institute, Royal Brisbane Hospital, Brisbane, Australia; ^bInstitute for Health and Biomedical Innovation, and School of Biomedical Science, Queensland University of Technology, Brisbane, Australia; ^cIstituto Zooprofilattico Sperimentale delle Venezie, Padua, Italy; ^dGeographic Information Systems, Department of Information Engineering, University of Padua, Padua, Italy

Table 2. CHIKV ‘La Reunion’ infection and dissemination to the wings/legs and saliva in *Ae. koreicus* mosquitoes maintained at 23 °C and fluctuating temperature (75 ± 5% relative humidity, 12 h light: 12 h dark cycle).

Treatment	Day post feeding	Engorged mosquitoes	Infection (tested)	Dissemination wings/legs (tested)	Dissemination saliva (tested)
23 °C	3	21	3 (21)	2 (3)	1 (3)
	10	19	3 (19)	2 (3)	1 (3)
	14	25	3 (25)	0 (3)	0 (3)
Fluctuating T °C	3	21	4 (21)	0 (4)	0 (4)
	10	21	0 (21)	NT*	NT*
	14	22	0 (22)	NT*	NT*

Note: NT* = Not tested as bodies were negative.



Article

Vector Competence of the Invasive Mosquito Species *Aedes koreicus* for Arboviruses and Interference with a Novel Insect Specific Virus

Stephanie Jansen ^{1,2}, Dániel Cadar ^{1,2} , Renke Lühken ¹ , Wolf Peter Pfitzner ³ , Hanna Jöst ¹, Sandra Oerther ^{4,5}, Michelle Helms ¹, Branka Zibrat ¹, Konstantin Kliemke ¹, Norbert Becker ^{4,6,7}, Olli Vapalahti ⁸ , Giada Rossini ⁹ and Anna Heitmann ^{1,*} 

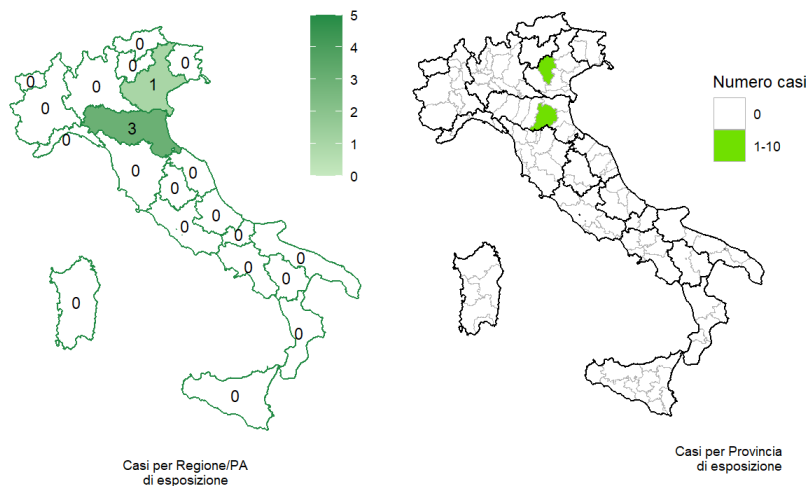
Table 1. Infection rates (IR), transmission rates (TR), transmission efficiency (TE) and body titer of *Aedes koreicus* at two different temperatures (14 days post-infection) for arbovirus infection with chikungunya Virus (CHIKV), West Nile Virus (WNV) or Zika Virus (ZIKV), and information about IR with Wiesbaden virus (WBDV) and coinfection rate (CR).

Arbovirus	Temperature °C	IR Arbovirus	TR Arbovirus	TE Arbovirus	Mean (95% Confidence Interval) log10 RNA Copies/Specimen	IR WBDV	CR
CHIKV	24 ± 5	17.6% (6/34)	0% (0/6)	0% (0/34)	1.37 (0.29–2.45)	8.8% (3/34)	8.8% (3/34)
CHIKV	27 ± 5	68.2% (103/151)	6.8% (7/103)	4.6% (7/151)	4.68 (4.15–5.22)	6.6% (10/151)	6.0% (9/151)
WNV	24 ± 5	90.6% (29/32)	0% (0/29)	0% (0/32)	7.09 (6.15–8.03)	3.1% (1/32)	3.1% (1/32)
WNV	27 ± 5	85.4% (35/41)	0% (0/35)	0% (0/41)	6.7 (5.72–7.68)	0% (0/41)	0% (0/41)
ZIKV	24 ± 5	78.9% (30/38)	0% (0/30)	0% (0/38)	4.97 (4.02–5.92)	0% (0/38)	0% (0/38)
ZIKV	27 ± 5	81.4% (35/43)	5.7% (2/35)	4.7% (2/43)	6.2 (5.19–7.21)	20.9% (9/43)	20.9% (9/43)

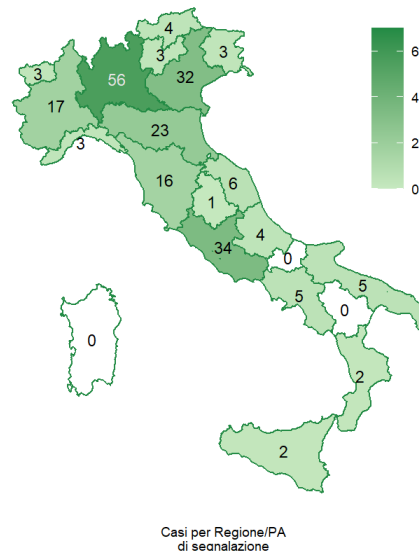


Dengue virus in Italy (2025)

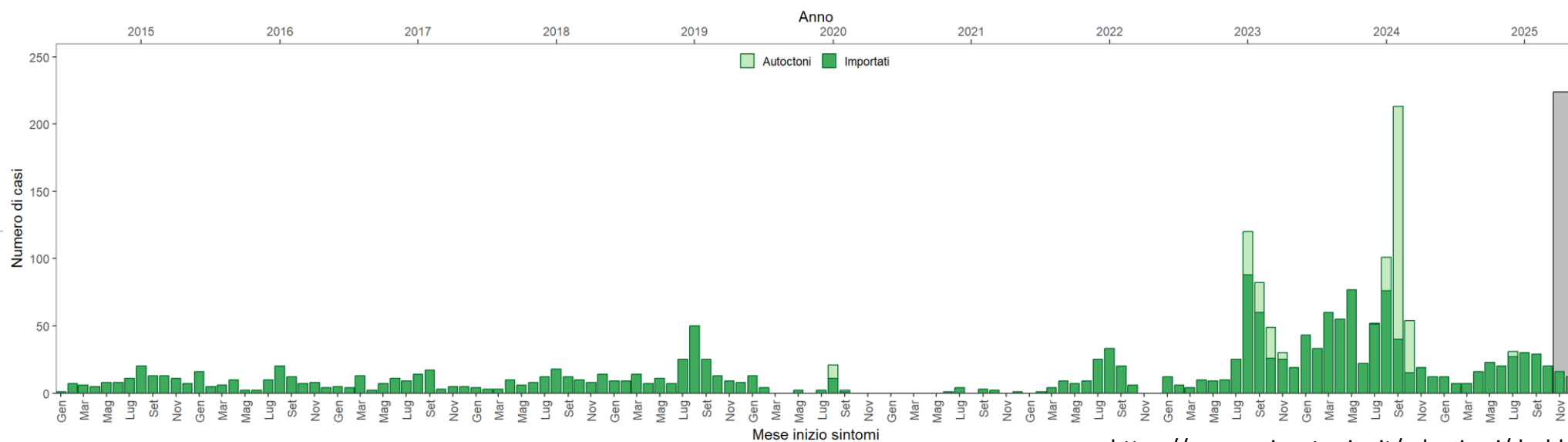
Casi autoctoni



Casi importati



223
Casi*
57% | 43%
Maschi | Femmine*
40 anni
Età mediana*
0
Decessi*
4 casi | 219 casi
Autoctoni | Importati*
*Dati in fase di consolidamento

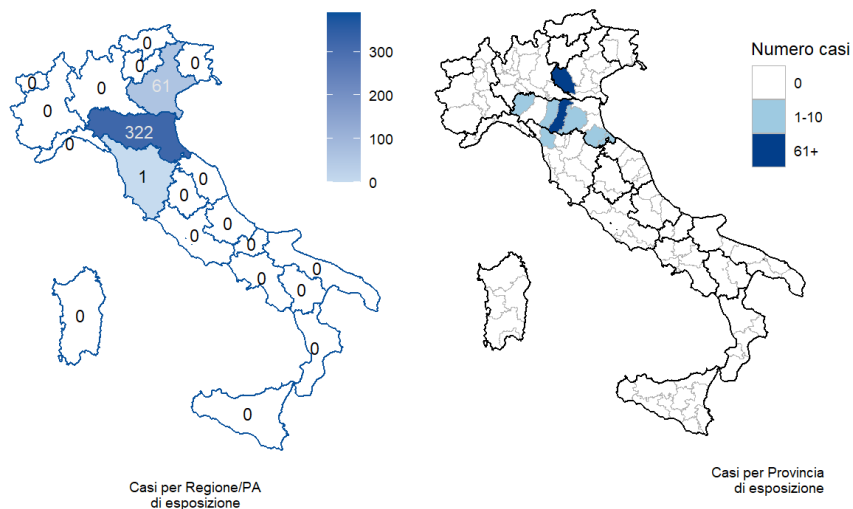


Il dato nell'area grigia, relativo agli ultimi due mesi, potrebbe essere sottostimato

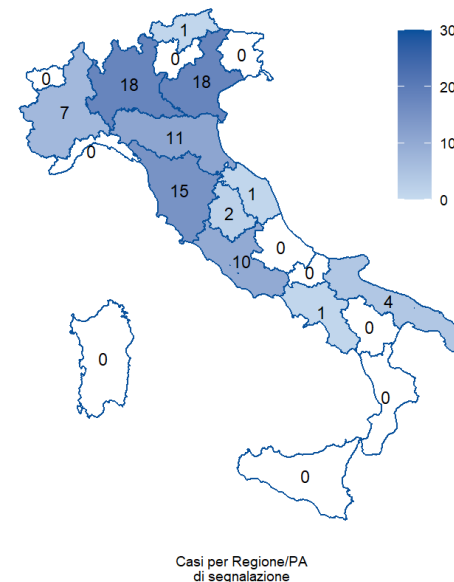
<https://www.epicentro.iss.it/arbovirosi/dashboard>

Chikungunya virus in Italy (2025)

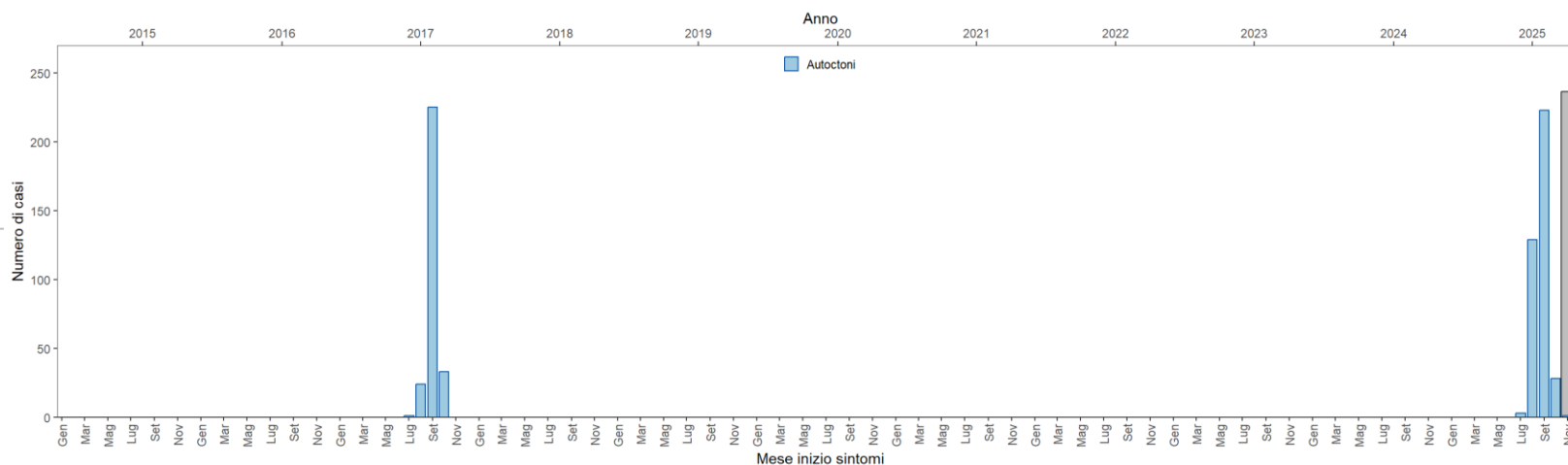
Casi autoctoni



Casi importati



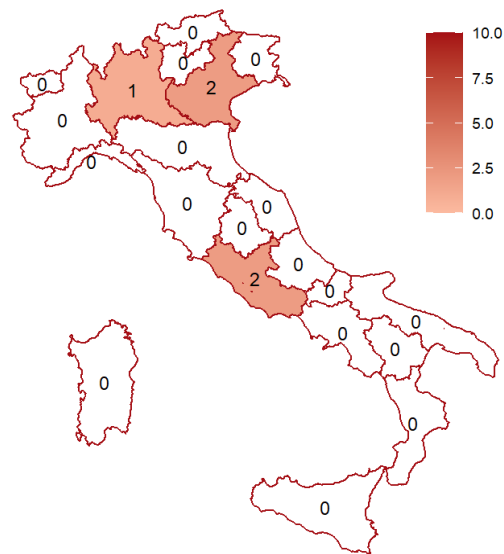
472
Casi*
50% | 50%
MaschiFemmine*
58.5 anni Età
mediana*
0 Decessi*
384 casi | 88 casi
Autoctoni | Importati*



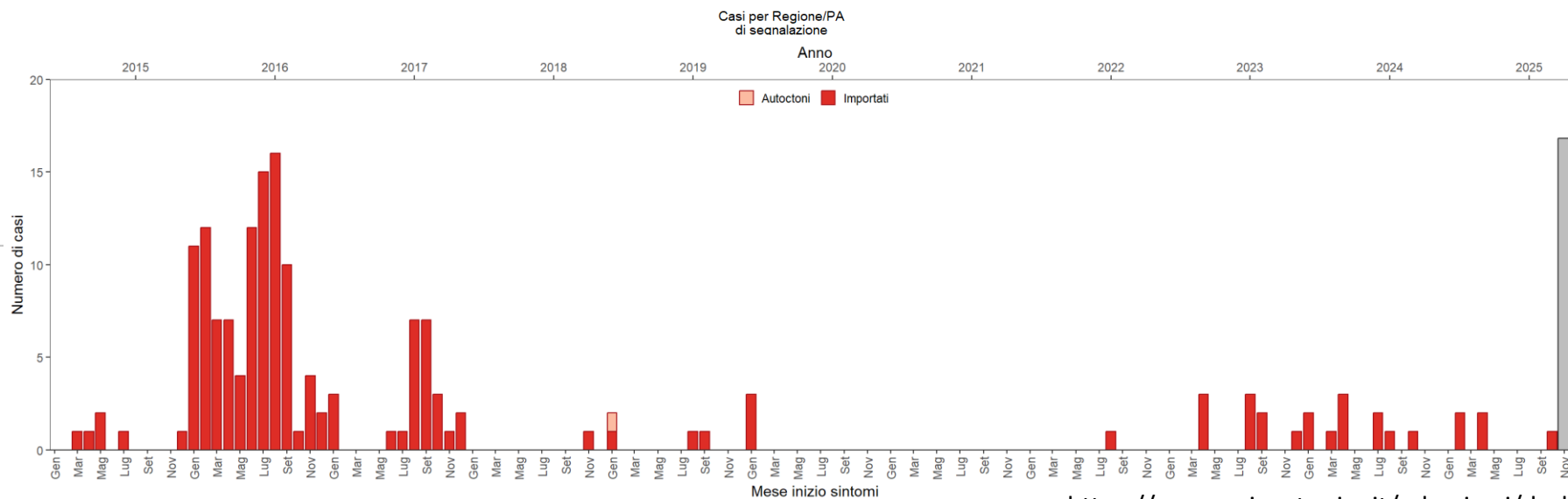
Il dato nell'area grigia, relativo agli ultimi due mesi, potrebbe essere sottostimato

Zika virus in Italy (2025)

Casi importati



5
Casi*
40% | 60%
Maschi | Femmine*
52 anni
Età mediana*
0
Decessi*
0 casi | 5 casi
Autoctoni | Importati*

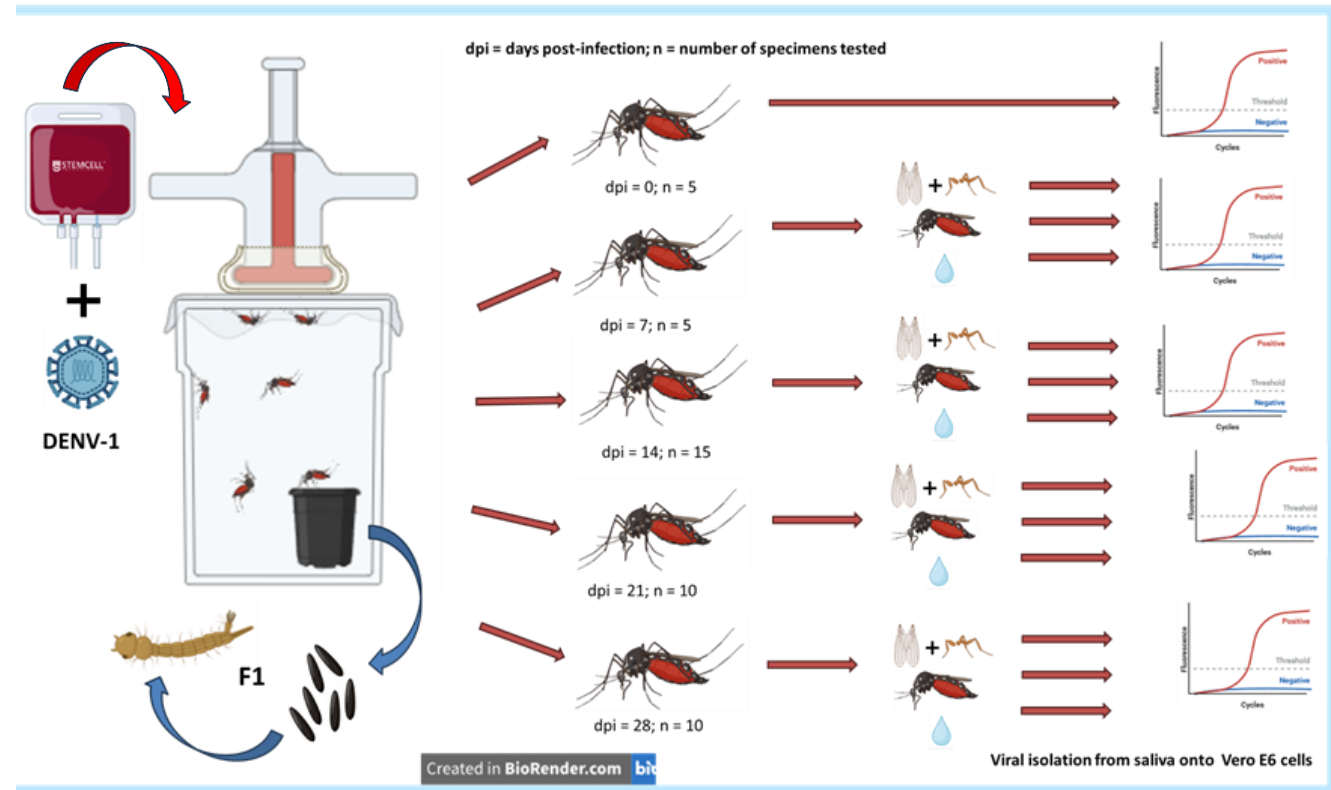


Il dato nell'area grigia, relativo agli ultimi due mesi, potrebbe essere sottostimato

Study of vector competence for DENV-1 in *Aedes koreicus*?

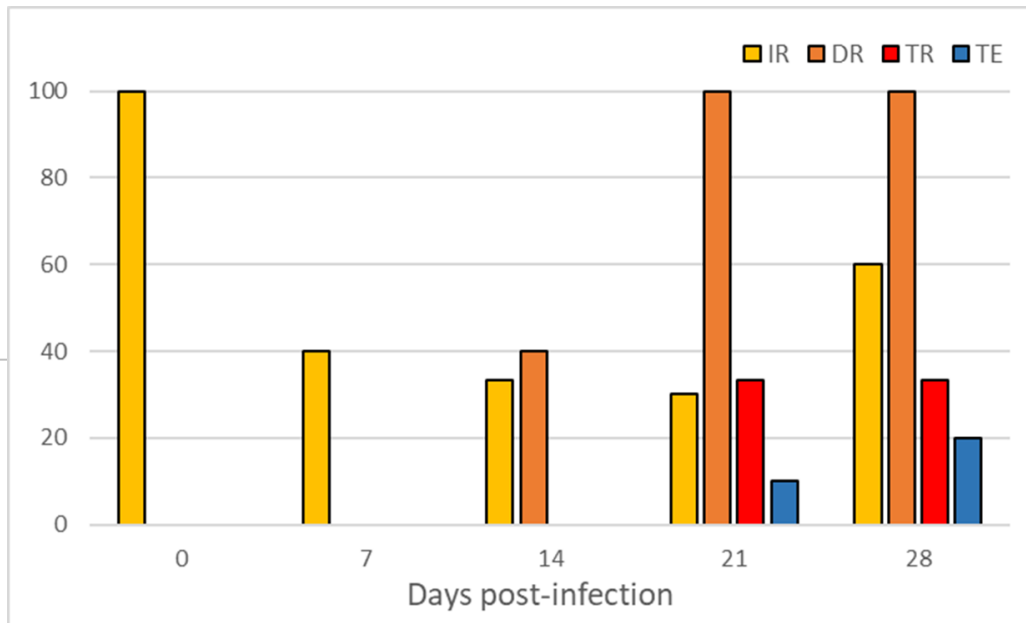
SPF rabbit whole blood in K₃EDTA diluted with DENV-1 virus. with a final titer of 1x10⁶ TCID₅₀/ml

- ❑ 26±1°C temperature
- ❑ 70% relative humidity, and a
- ❑ 12:12h light: dark cycle
- ❑ supplied with a 10% sucrose solution ad libitum



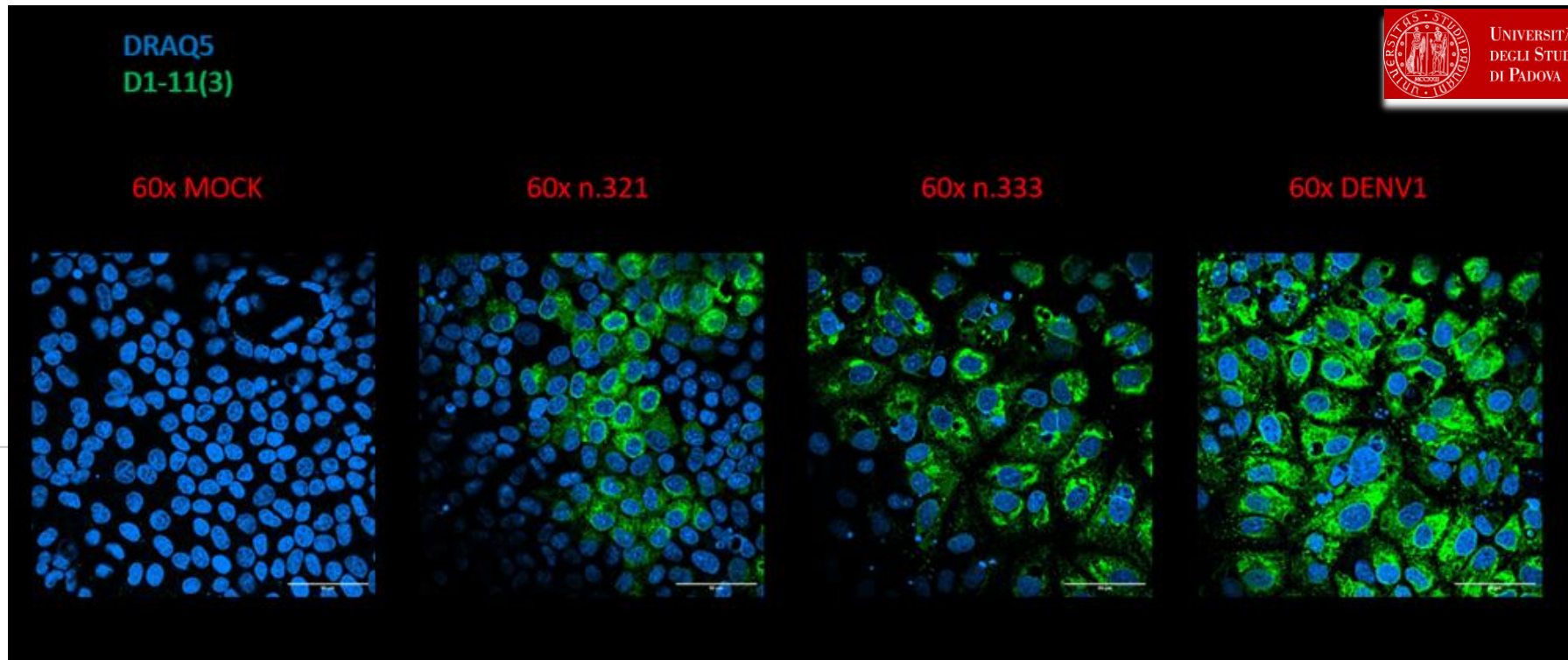


Days post infection	IR	Main Ct value (range)	DR	Main Ct value (range)	TR	Main Ct value (range)	TE
0	100% (5/5)	-	-	-	-	-	-
7	40% (2/5)	24.83 (22.46-27.02)	0% (0/2)	-	-	-	-
14	33.3% (5/15)	24.82 (22.06-28.66)	40% (2/5)	25.64 (25.13-26.15)	0% (0/2)	-	0% (0/15)
21	30% (3/10)	22.99 (22.50-23.27)	33.3% (1/3)	28.16 (27.80-28.87)	33.3% (1/3)	37.16	10% (1/10)
28	60% (6/10)	22.21 (18.17-25.02)	33.3% (2/6)	31.42 (20.86-42.32)	33.3% (2/6)	31.83 (30.46-33.20)	20% (2/10)



- ❑ DENV-1 was first detected in saliva at 21 dpi and again at 28 dpi highlighting a **long extrinsic incubation period**

Three positive saliva were found but only two were isolated onto Vero E6 cell monolayers showing cytopathic effect (CPE)



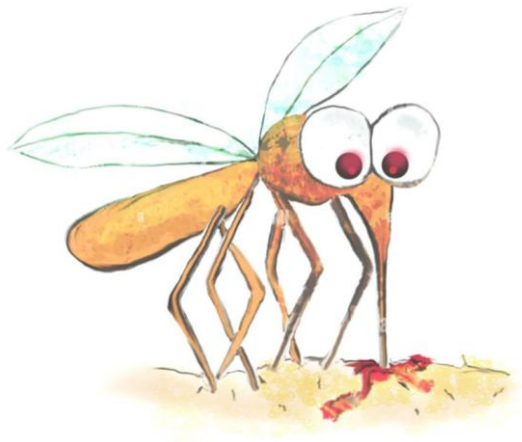
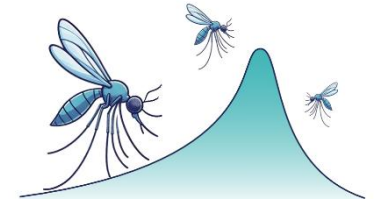
Final remarks

- ❑ *Aedes koreicus* is well established in Europe
- ❑ Knowledge of its vector competence for emerging and re-emerging arboviruses infection are essential to assess correctly the risk of pathogen transmission to human population
- ❑ It is urgent the need to have standardized guidelines for researchers for robustness and comparison of experimental results

PREPARATION

“KNOW YOUR ENEMY & KNOW YOURSELF”





Thank you All for attention!

Federica Gobbo, DVM

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CRN per la ricerca scientifica sulle malattie infettive nell'interfaccia uomo-animale

WOAH Collaborating Centre for Diseases at the Animal/Human Interface

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